

Modelling the thermal degradation of potassium alum using artificial neuronal networks

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Abstract

This work assesses the use of artificial neuronal networks as a modelling tool for the thermal degradation of potassium alum. It was executed using the experimental data from the thermal gravimetric analyses, with $10\text{ }^{\circ}\text{C min}^{-1}$ heating rate, with and without the presence of charcoal as a reducing agent. The modelling process evaluated the benefits of different numbers of neurons, hidden layers, input variables, different activation functions and training algorithms. The parameters used to evaluate each scenario were the Sum of Square Errors (SSE) and the Determination Coefficient (R^2). The model that presented the best result for the $10\text{ }^{\circ}\text{C min}^{-1}$ heating rate, without the presence of charcoal, used temperature as an input model variable, while the mass loss was used as an output variable. About the network's architecture, 1 hidden layer and 10 neurons were employed. The algorithm *trainlm* was used as training algorithm and *logsig* as activation function. For the $10\text{ }^{\circ}\text{C min}^{-1}$ heating rate, with the presence of charcoal, it was temperature as an input, mass loss as output, with 2 hidden layers, 5 neurons each, *tansig* as activation function in both layers and *trainlm* as training algorithm. Both cases were chosen for presenting the best results, without unnecessary computational expense. The overall results showed the great capacity of the neuronal networks for modelling, especially for the capture of nonlinear behavior since the tested models presented really adequate fits to the experimental data considered.

1. Introduction

With the constant population growth, agriculture and its development have been considered a worldwide concern and, therefore, techniques to ensure good yields have been studied extensively. One of the most important ions for plant nutrition is potassium, with fertilizers consuming most of its production. Although most of the potassium is usually used in the form of KCl (potassium chloride), some specific cultures are sensitive to chlorine or need sulfur ions, so the best alternative is the use of K_2SO_4 (potassium sulfate). Given the importance of this ion, it is clear the relevance of finding new routes to its production, especially in a country like Brazil, where the economy is heavily based on the agriculture, with the country being the key market of several food related products. However, the internal production of this ion is far from meeting its own demand. In 2013, Brazil's internal production of KCl was able to supply only 6.03 % of its own consumption and, in the same year, the country imported 40.92 thousand tons of K_2SO_4 (Oliveira, 2014).

Emerging as a prominent alternative is the use of glauconite, a mineral found abundantly in Brazil. Glauconite is known to be a mineral with a significant presence of potassium (McRae, 1972) and the potential of this mineral has been explored by studying the possible routes to get to the potassium ion, in the form of K_2SO_4 (Brocchi et al., 2015). One possible route studied involves the calcination of the glauconite, then sulfuric acid digestion, followed by a few unit operations (Nofal et al., 2019). In this process, $KAl(SO_4)_2$ (potassium alum) is one intermediate product of relevance for the obtainment of the potassium sulfate.

The potassium alum can usually be found in its hydrated form $KAl(SO_4)_2 \cdot 12H_2O$ and through thermal decomposition can lead to the formation of K_2SO_4 and Al_2O_3 (aluminum oxide). Aluminum is the second most used metal in the world, with its oxide being used in different industrial sectors and it is usually obtained through the Bayer process of beneficiation of the bauxite (Liu and Müller, 2012). Unfortunately, the Bayer process has a toxic residue, known as red mud, that can affect water quality and contaminate the soil (Liu and Naidu, 2014). Given these hazards, the use of the glauconite presents itself even more as an advantageous alternative.

The viability of the process has been already explored by other studies (Nofal et al., 2019; Souza et al., 2019) that helped establish the treatment route of the glauconite and analyzed the effects of reducing agents, but the kinetic mechanism is still uncertain. Nevertheless, these studies developed a proposed mechanism that suggest the decomposition of the potassium alum happens in a series of chain reactions, being 4 for the loss of the water molecules and 2 to obtain the 2 desired products. In order to assist the development and enable the implementation of the process at industrial scale it is important to comprehend more about this process, know the proper amount of reagents, the optimal experimental conditions, avoiding waste of materials and saving energy, without compromising the security of the process.

The thermal gravimetric analyses (TGA) are laboratorial scale analyzes that can reproduce the conditions of the thermal decomposition of materials (Jain et al., 2016), being ideal for the study of the decomposition of the potassium alum into potassium sulfate. To further help the implementation of the process it's important to model the results of the TGA as to understand the behavior of the compound throughout this process, test the suggested mechanisms, design the scaling of possible projects involving the process and even test safety and health conditions applied to it.

There are a few possible ways available of modelling TGA results. One of them is through graphic methods (Aboulkas et al., 2010; Bianchi et al., 2008; Huang et al., 2016; Jain et al., 2016; Lim et al., 2016; Mishra and Mohanty, 2018; Ozawa, 1965), but those have the disadvantage of not using the whole

data available by the analysis, possibly distorting the statistical interpretation (Toniolo et al., 2008). Another option is through estimation of parameters using the mathematical model derived by the mass balance of the species present in the system, which can provide good models capable of representing well the experimental data (Nofal et al., 2019). However, this approach requires complex calculations, justified by the large number of parameters to estimate for the majority of the cases. Another disadvantage of this method is the need for previous knowledge of the process and the need for assumptions of its functioning. In the specific case of the modelling via mass balance of the potassium alum's TGA results developed by (Nofal et al., 2019), the intermediary products are assumed to exist, but the authors could not prove their existence by experimental results, since XRD (X-ray diffraction) was done only for samples collected at the end of the experimental test.

As a counterproposal, artificial neuronal networks (ANN) can be seen as an alternative to the problems that the other modelling pathways may present. The ANN uses a mathematical simulation based on neurological systems that can process extensive database. Through different numbers of layers, training algorithms, activation functions and number of neurons, it can provide effective results in quality models, with high capacity of learning patterns without needing any previous knowledge of the process functional relationship (Lecun et al., 2015).

The neuronal networks are already being used for modelling different types of processes (Fidalgo et al., 2018; Liu et al., 2018; Mu et al., 2004; Pastukhov, 2015; Rego et al., 2018; Valim et al., 2017). These studies also evaluate the neuronal networks itself, by analyzing the advantages of the use of more hidden layers (Valim et al., 2017), of larger number of neurons in the layers (Fidalgo et al., 2018) and of different activation functions (Rego et al., 2018).

The ANN represents a simple method, that can model systems with different behaviors, without the need to simplify them (Bach and Poggio, 2016). It is a great way to study the process availability at bigger scales and it can also facilitate the process of online monitoring, since it provides simple equations that can predict the steps of the process.

The neuronal networks not only can process large amounts of data, but they in fact prefer them, because it increases the accuracy of the results (Bach and Poggio, 2016; Krizhevsky et al., 2017). Because of that, the neuronal networks can be a great method for modelling TGA results, that typically accumulates a big dataset. As of this moment, there are a couple of works using the ANN to model TGA results of different compounds (Chen et al., 2017; Conesa et al., 2004; Fakhim et al., 2013; Monteiro et al., 2018; Naqvi et al., 2018; Yildiz et al., 2016), and they provided good results, being successful and

referenced work achieved general good results. The only case in which the model predictions differed most from the experimental data was the experimental condition that applied $10\text{ }^{\circ}\text{C min}^{-1}$ as heating rate, being the reason why the present study chose this specific case (with and without the presence of charcoal) as the main focus of investigation.

(Pysiak and Glinka, 1981) indicates that the initial mass loss occurring is due to the water loss. As the main focus is on the thermal decomposition of the potassium alum itself, this initial part of the data was excluded from the current modelling, with the new start point being defined at 25 min, as illustrated in Figure 3.

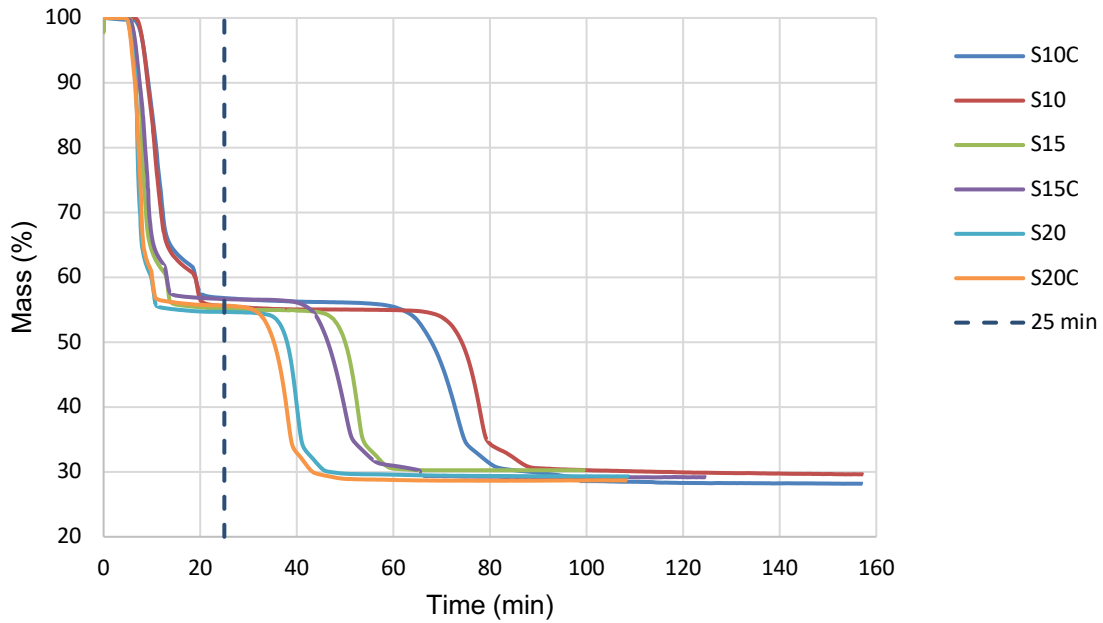


Figure 3: Graphic of the mass loss over time of the 3 heating rates, with and without charcoal

Table 1 shows the meaning of the data series indicated in Figure 3.

Table 1: Information assignment for the legend on Figure 3

S10	Heating rate of $10\text{ }^{\circ}\text{C min}^{-1}$ without the presence of charcoal
S10C	Heating rate of $10\text{ }^{\circ}\text{C min}^{-1}$ with the presence of charcoal
S15	Heating rate of $15\text{ }^{\circ}\text{C min}^{-1}$ without the presence of charcoal
S15C	Heating rate of $15\text{ }^{\circ}\text{C min}^{-1}$ with the presence of charcoal
S20	Heating rate of $20\text{ }^{\circ}\text{C min}^{-1}$ without the presence of charcoal
S20C	Heating rate of $20\text{ }^{\circ}\text{C min}^{-1}$ with the presence of charcoal

The process of construction of the neuronal network requires a sequential path of steps. Initially the data needs to be separated into 2 groups (training and testing) and normalized, then it is decided which variables will be used as an input (meaning, which ones will enter the neuronal network for calculation) and which will be used as output (meaning, which ones will be the

expected result of the model). The next step is to select the number of hidden layers (what will imply in more or less internal steps that the data will go through in the process of obtaining the result), the number of neurons in each layer (that implies in the number of points in which the data will go through calculations), the activation function (that will determine what will be the calculations made) and the training algorithm (that will determine how the model will be updated). After this process, it is necessary to evaluate the results given by the model and judge whether the model is suited to the experimental data and if not, repeat the process with different parameters. Figure 4 shows, in a generic way, those steps of the process.

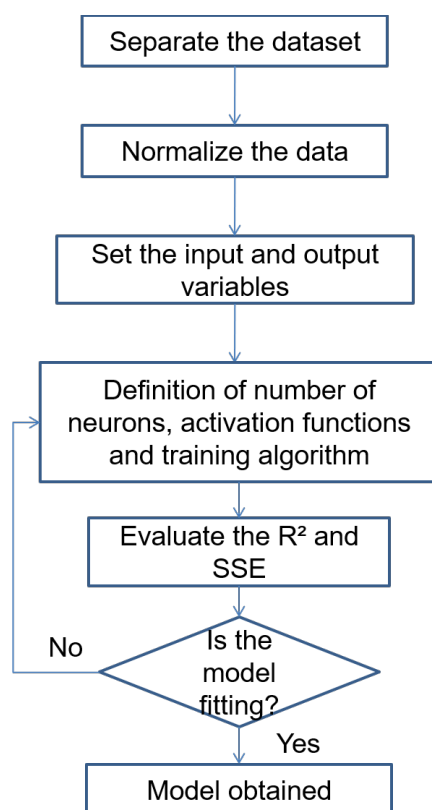


Figure 4: Flowchart of the process of achieving a model via ANN

The division of the data groups in this work was made taking into consideration the fact that the training process requires a larger dataset, as to enable the optimization of the network. Thus, 75 % of the data (1988 points) went to the training group and the remaining 25 % (662 points) went to the testing. As the experimental data for the 10 °C min⁻¹ heating rate with and without charcoal was the same size, the division was the same for both cases.

The software used was Matlab R2019a and the chosen technique was of supervised training where, to each model made, it was chosen input and output data, so the training process would self-evaluate by comparison (Valim et al., 2017). By the end of the training, the algorithm runs the obtained model by the

testing dataset and returns, as an analysis of its performance, the Sum of Square Errors (SSE) and the Determination Coefficient (R^2). The value of SSE closer to 0 and the value of R^2 closer to 1 indicates the model obtained good adjustment and can be used to predict the loss mass in the TGA results. The Equations (1) and (2) show the mathematical formulation for SSE and R^2 , respectively. Besides that, it was also taken into consideration which topologies could present fitting results with less neurons, since the more neurons used, the greater the computational expense.

$$SSE = \sum_{i=1}^n (y_{observed} - y_{predicted})^2 \quad (1)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_{observed} - x_{predicted})^2}{\sum_{i=1}^n (y_{observed} - y_{mean})^2} \quad (2)$$

The construction of the neuronal network depends on the determination of two important factors: the training algorithm and the activation function. The training algorithms are based on variations of the Backpropagation algorithm (Graupe, 2007), in which information goes forward from the input layer to the output layer. Then, the error is calculated and the information goes backward in the network updating the weights and bias values in order to minimize the error value (Rego et al., 2018). The weights are proportional values to each information in the neurons that multiply the variables and the bias are independent parameters that are added to the information before it is directed to the activation function (Valim et al., 2017). Different algorithms can be used to update the weights and bias. The ones used in this work were *trainlm* (Levenberg-Marquadt), *trainbr* (Bayesian regularization), *trainoss* (one step secant), *traingdx* (gradient descendend momentum) and *traincgb* (conjugate gradient with Power-Beale) (Gitifar et al., 2013). The activation function are ways of processing the information in each neuron, calculating an output using the weights and bias (Valim et al., 2017). The ones used in this work are hyperbolic tangent, logistic and linear. In the Matlab software, these functions are called *tansig*, *logsig* and *purelin* and they are represented in Equations (3) to (5).

$$tansig(x) = \frac{2}{1 + e^{-2x}} - 1 \quad (3)$$

$$logsig(x) = \frac{1}{1 + e^{-x}} \quad (4)$$

$$purelin(x) = x \quad (5)$$

Other determining factor is the structure of the network, which consists in the input layer (that should have the same number of neurons as of variables provided), one or more hidden layers (where the *tansig* and *logsig* functions are used) and one output layer (that should have the same number of neurons as of desired responses), where one linear activation function, *purelin*, is applied. In this structure, the number of neurons in the hidden layers is open to choose, since the convention adopted was to use an amount equal to or greater than the number of neurons in the input layer. The number of modifiable parameters should be smaller than the group of data used for training. Thus, no restriction as to adding neurons was significantly established, as the dataset used was extensive. Therefore, different numbers of neurons were tested along with networks with 1 and 2 hidden layers. Figures 5 (a) and 5 (b) show a generic scheme of the flow of the data through a neuronal network with two input variables with 1 and 2 hidden layers, respectively.

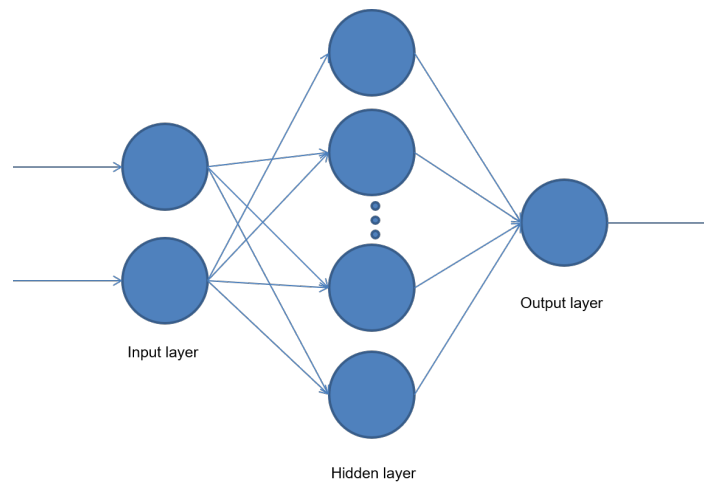


Figure 5 (a): Route of the data throw a neuronal network with 2 input variables and 1 hidden layer

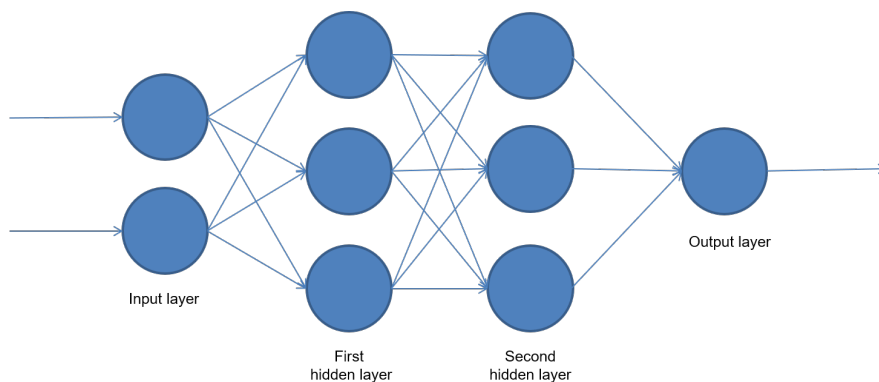


Figure 5 (b): Route of the data throw a neuronal network with 2 input variables and 2 hidden layers

In this work, considering all the variety of factors previously established, it was tested the modelling of the potassium alum's TGA results using different variables of input and output. The dataset from the experiment provided the variations, through time, of temperature, percentage of mass loss and DTG (derivate of the thermogravimetry), the first derivate of the curve of the mass loss as a function of time. Therefore, the models developed in this work are illustrated in Table 2.

Table 2: Models tested for different topologies in this work

Model	Presence of charcoal	Input Variables	Output Variables
1	No	Temperature	Mass loss
2	No	Temperature and DTG	Mass loss
3	Yes	Temperature	Mass loss
4	Yes	Temperature and DTG	Mass loss

3. Results and Discussion

In each model, several topologies were evaluated, with 1 and 2 hidden layers tested and varying the number of neurons, the training algorithm and the activation function. Table 3 displays a listing of some of the relevant results obtained in the 4 models, highlighting the ones considerate the best results for each case.

Table 3: Most relevant topologies for each model tested and its characteristics

Model	Training Algorithm	Neurons Hidden Layer 1	Activation Function Hidden Layer 1	Neurons Hidden Layer 2	Activation Function Hidden Layer 2	SSE	R ²
1	trainlm	10	logsig	-	-	0.0436	0.99992
	trainbr	10	logsig	-	-	0.0497	0.99993
	trainoss	10	logsig	-	-	0.3390	0.99939
	traingdx	10	logsig	-	-	2.0100	0.99636
	traincgb	10	logsig	-	-	0.1410	0.99974
	trainlm	10	logsig	10	logsig	0.0246	0.99996
2	trainlm	10	tansig	-	-	0.0475	0.99992
	traingdx	10	tansig	-	-	3.9100	0.99302
	traingdx	10	logsig	-	-	21.3000	0.96135
	trainlm	30	tansig	-	-	0.0377	0.99993
	trainlm	5	tansig	5	tansig	0.0411	0.99993
	trainlm	10	logsig	10	tansig	0.0429	0.99993
3	trainlm	5	tansig	5	tansig	0.0072	0.99998
	trainlm	5	logsig	5	logsig	0.0271	0.99995
	trainoss	10	tansig	-	-	0.0540	0.99989
	trainoss	10	logsig	-	-	0.0810	0.99984
	trainbr	5	logsig	5	logsig	0.0165	0.99997
	trainbr	5	tansig	5	tansig	0.0169	0.99998
4	trainbr	5	tansig	5	tansig	0.0068	0.99999
	trainbr	5	logsig	-	-	0.0272	0.99995

	trainbr	10	logsig	-	-	0.0277	0.99996
	trainoss	10	tansig	-	-	1.9200	0.99664
	trainoss	10	logsig	-	-	2.2800	0.99634
	trainbr	3	tansig	3	tansig	0.0320	0.99995

In Figures 6 and 7, we can see the graphics comparing the data with the results of the best topology found for the heating rate $10\text{ }^{\circ}\text{C min}^{-1}$ without charcoal. For the choice of the best model for this case, it was analyzed the best topologies for models 1 and 2, taking into consideration the advantages of each situation. The best result was the model 1, with 1 input variable, and its topology of 1 hidden layer with 10 neurons in it, *logsig* as the activation function and *trainlm* as the training algorithm and this model resulted in R^2 of 0.99992 and SSE of 0.0436.

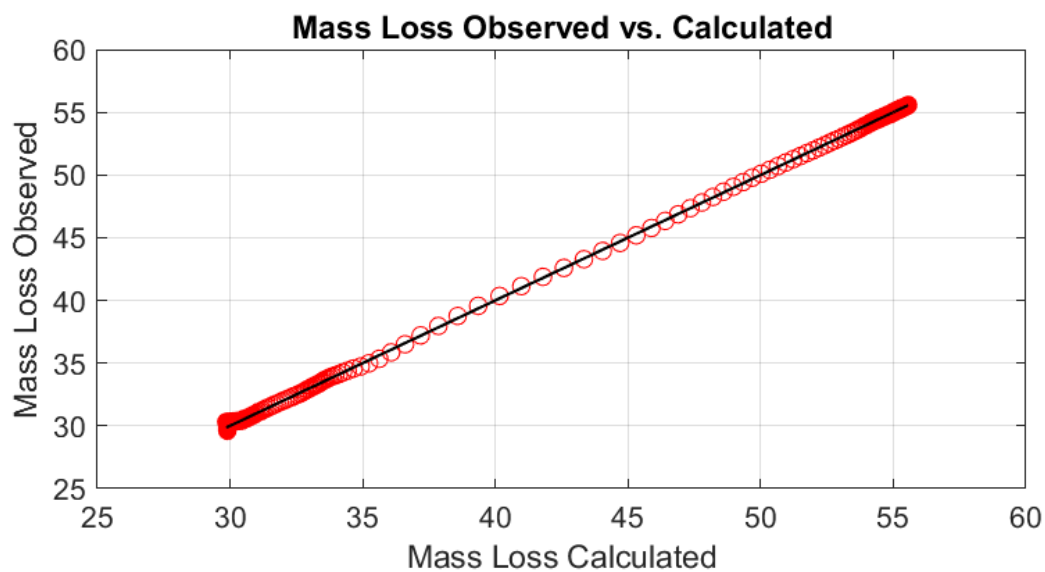


Figure 6: Graphic of the mass loss (%) of the data vs. the mass loss obtained by the topology of the $10\text{ }^{\circ}\text{C min}^{-1}$ heating rate without charcoal, with 1 input layer, 10 neurons, *logsig* as activation function and *trainlm* as training algorithm

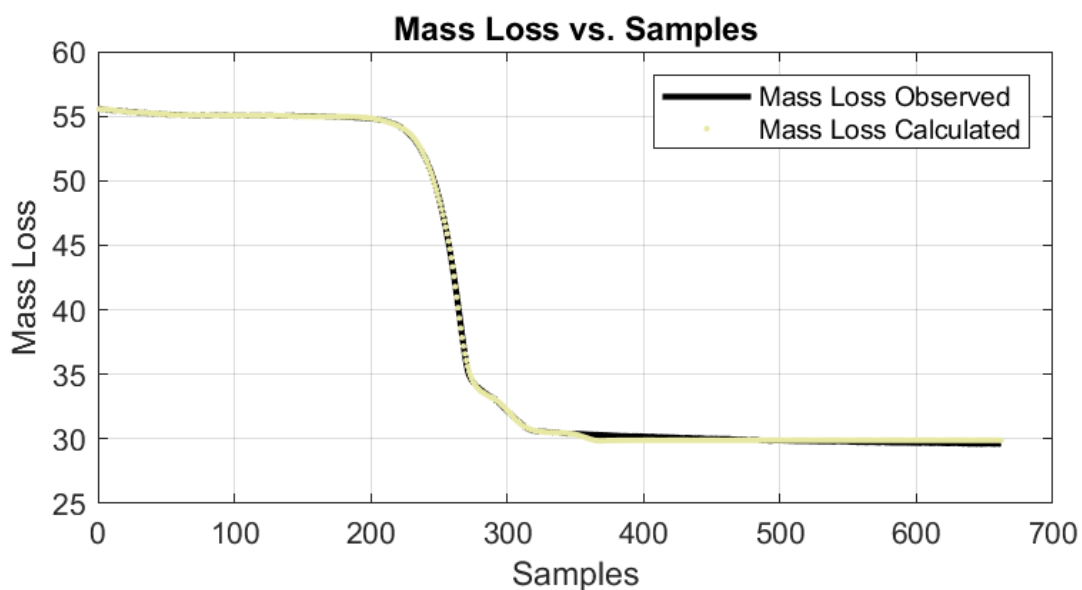


Figure 7: Graphic of the mass loss (%) of the available data for each analysis and the corresponding results of the 10 °C min⁻¹ heating rate without charcoal, with 1 input layer, 10 neurons, *logsig* as activation function and *trainlm* as training algorithm

In Figures 8 and 9, we can see the graphics comparing the data with the results of the best topology found for the heating rate 10 °C min⁻¹ with charcoal. For the choice of the best model for this case it was executed the same process made for models 1 and 2, but comparing now models 3 and 4. The best result was the model 3, with 1 input variable, and its topology of 2 hidden layers, both with 5 neurons and *tansig* as the activation function and *trainlm* as the training algorithm and this model resulted in R² of 0.99998 and SSE of 0.0072.

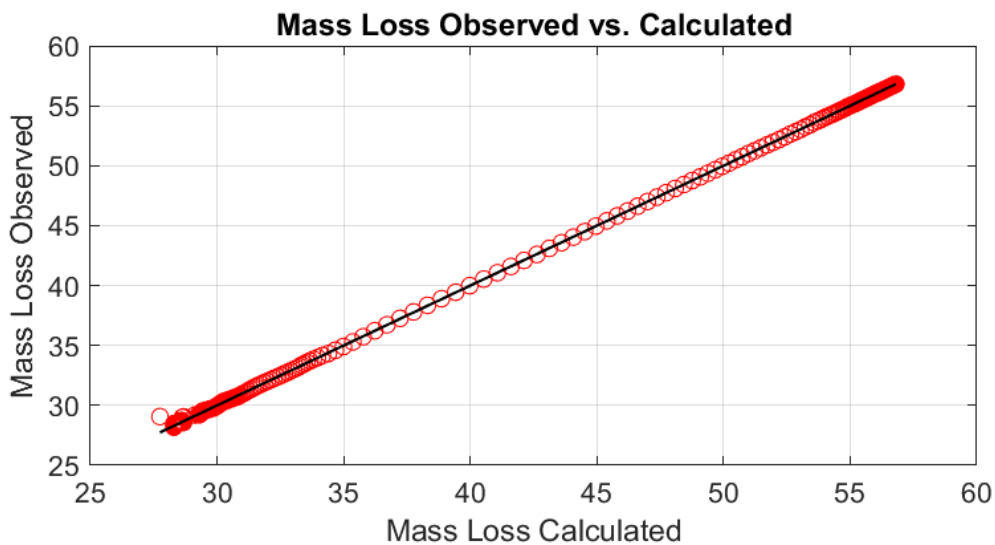


Figure 8: Graphic of the mass loss (%) of the data vs. the mass loss obtained by the topology of the 10 °C min⁻¹ heating rate with charcoal, with 2 input layers, 5 neurons each, *tansig* as activation function for both and *trainlm* as training algorithm

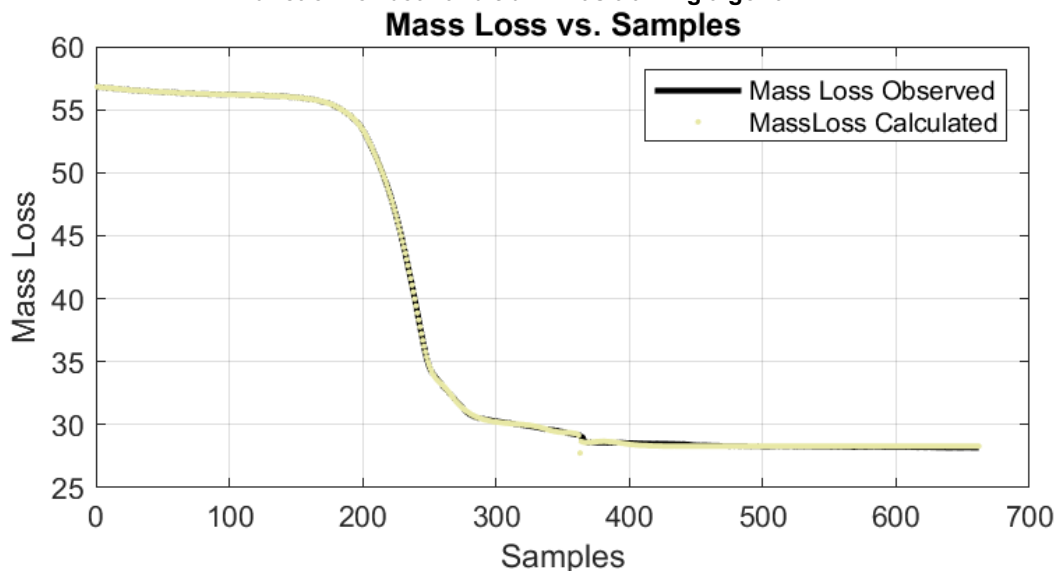


Figure 9: Graphic of the mass loss (%) of the available data for each analysis and the corresponding results of the 10 °C min⁻¹ heating rate with charcoal, with 2 input layers, 5 neurons each, *tansig* as activation function for both and *trainlm* as training algorithm

The Tables 4 and 5 show the weights and bias of the two models, chosen as the best, presented as a result. Those parameters enable the quick calculation of a desired mass loss, given the temperature of the process at the moment. This calculation can be made through the use of Equation 5, where w is the weight, b is the bias, x is the temperature and y is the mass loss obtained at each temperature.

Table 4: Weights and bias of the topology of the 10°Cmin⁻¹ heating rate without charcoal, with 1 input layer, 10 neurons, *logsig* as activation function and *trainlm* as training algorithm

j	Weights		Bias	
	Input Layer	Hidden Layer	Input Layer	Hidden Layer
1	31.0197	-0.0741	-29.5607	-0.1556
2	-29.0202	0.2208	18.9633	-
3	24.9302	0.0138	-22.8158	-
4	46.0803	-0.7776	-20.3517	-
5	-20.7458	0.8545	7.5354	-
6	-26.4373	0.0333	3.7476	-
7	-28.6552	0.0065	-6.5060	-
8	-27.7250	0.0019	-16.0370	-
9	-28.0510	0.0157	-21.3497	-
10	-28.8919	0.0243	-26.8604	-

Table 5: Weights and bias of the topology of the 10°Cmin⁻¹ heating rate with charcoal, with 2 input layers, 5 neurons each, *tansig* as activation function for both and *trainlm* as training algorithm

j	Weights							Bias		
	Input Layer	Hidden Layer 1					Hidden Layer 2	Input Layer	Hidden Layer 1	Hidden Layer 2
1	13.0746	24.8766	-0.5464	0.9860	-1.0002	-1.1952	-0.2227	-12.8833	-1.6027	0.0052
2	-12.5559	-9.0075	4.8639	-1.6268	0.5500	-3.2874	1.1900	1.0414	-0.7067	-
3	5.5487	-0.9959	-1.7366	2.9763	2.3277	-0.2380	0.4608	-2.1499	-0.1412	-
4	-3.7292	-0.2176	-2.3185	1.4144	0.8874	4.0350	-0.2054	-1.4004	1.7416	-
5	6.5936	-31.5366	0.2316	1.6878	-1.1267	-4.0596	-0.2047	6.3463	-12.8870	-

$$y = \sum_{j=1}^n \left(\sum_{i=1}^m w_{ji} x \right) + b_j \quad (6)$$

Evaluating the results for model 1, it's possible to see differences, even with the use of the same activation function and the same number of neurons. Both *trainlm* and *trainbr* showed significant better results, with 1 or 2 hidden layers. The other training algorithms provided similar SSE and R^2 , with the *traingdx* being relevantly worse, especially considering the SSE. Also, in the variation of *trainlm* with 1 and 2 hidden layers, the results were not significantly different, thus not making the extra computational expense worth the use of another hidden layer, with more neurons.

In the results obtained for model 2, the two attempts of using *traingdx* (with different activation functions) highlighted, displayed SSE and R^2 far worse than all the other training algorithms, as it did in the case of model 1. This consolidated the pattern that was seen throughout the process of developing the models, that this training algorithm was not a good fit for the analysis of this work.

Also, in model 2 the same training algorithm and activation function was tested using 10 and 30 neurons, but the results were statistically the same, showcasing that the increase of neurons doesn't always have an impact on the model (this can also be seen with two hidden layers). This fact impacts directly in the decision-making of which model to use, since the more neurons it uses, the longer the processing is for the neuronal network and the more it requires of the machine being used. In this study the computational expense was not a defining point, since even with larger amounts of neurons the processing time was not significant, but in cases with larger data this could be very impactful, not only in time expend, but in the machinery capacity.

For model 3, the best performing algorithms, *trainlm* and *trainbr*, were tested with the same number of neurons and with two layers, changing the activation functions to evaluate if there was a significant impact. With *trainbr* the two scenarios resulted in practically equal results, but with *trainlm* the use of the *logsig* function in the hidden layers resulted in a better SSE than the use of *tansig*, demonstrating the variation that can occur specific to the interaction of each algorithm and function with the data.

In the model 4, the *trainoss* algorithm was tested with both activation functions and gave SSE and R^2 values worse than most of the others attempts. Comparatively, in model 3, *trainoss* was tested with the same number of neurons and it displayed good results. Even though these results were not the best of model 3, it showcases the relevance of testing an algorithm even if it displays bad results in one specific topology, since it doesn't necessarily imply in it not being appropriate for others.

In all the four models tested, the best topologies were chosen not only because they represented the best fit to the data, with the best SSE and R^2 , but because they were the ones that performed the best without unnecessary

additions (like more neurons or another hidden layer when not significantly impactful for the final result). Besides that, it was also made a comparison within the models. Since the models 2 and 4 were the ones with two variables as an input, they both implied in extra computational time consumption. So, as those models presented very similar results to those of models 1 and 3, the use of an extra input variable did not demonstrate itself as valuable.

In (Nofal et al., 2019), it's possible to see that the phenomenological model obtained showed in general a good fitting for the experimental data, coinciding with the dataset for the most part. But it's also possible to see that in some points the model results distain from the data and, in comparison with Figures 6 and 8, of the best results achieved with the neuronal networks, it is clear that the ANN gave a best fitted model to the data in both cases (with and without the presence of charcoal). Furthermore, the ANN's models required much less complicated calculations, resulting in a quicker and simpler process of achieving even better results. Alongside these advantages, differently from the phenomenological process, the neuronal network modelling did not need the assumption that the decomposition of the potassium alum occurred according to the mechanism illustrated in Figure 1. Even though the proximity of the model obtained by (Nofal et al., 2019) with the data of the TGA indicates that the proposed mechanism can be correct, there is still a possibility of it being partially incorrect. But, since the ANN does not require any supposition about the kinetic mechanism, those uncertainties do not impact the final model.

4. Conclusion

The use of the artificial neuronal network for modelling of the potassium alum's TGA results was proven to be very efficient. The models chosen as the best fit to the data were: temperature as an input, with mass loss as an output, 1 hidden layer, with 10 neurons, *trainlm* as training algorithm and *logsig* as activation function (for the 10 °C min⁻¹ heating rate, without charcoal); and temperature as an input, with mass loss as an output, using 2 hidden layers, with 5 neurons each, *tansig* as activation function in both layers and *trainlm* as training algorithm (for the 10 °C min⁻¹ heating rate, with the presence of charcoal). The evaluation of the process of construction of the network highlighted *trainlm* and *trainbr* as better fitting algorithms to this data, but also showed the relevance of testing different algorithms, since they can result in different responses for different data. This work also demonstrated the capacity of the ANN to provide fitting models with a quick and simple process, with the results been potentially even more representative of the data than other known modelling methods, especially in cases of nonlinear behavior.

5. References

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